Roll No.....

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

DEPARTMENT OF CHEMISTRY

END SEMESTER EXAMINATION, Nov/Dec 2025

Question Paper

Programme: **B. Tech. 1st Year**
Semester: **I**
Subject: **Energy and Environmental Science**

Maximum marks: **50**
Time allowed: **03 hours**
Subject code: **CHIC101**

Note:

1. Before starting to write the solutions, candidates should check the Question Paper for any discrepancies and also ensure that they have been delivered the Question Paper of the right course code and the subject.
2. Justify your answers with suitable examples, chemical reactions, and neat labelled diagrams, wherever required.
3. All questions carry equal marks. Attempt **any five questions**.

Q. No.	Description	Marks
1	a) Define biodiversity and explain its types. Describe various factors responsible for the loss of biodiversity. b) Explain the model of energy flow in an ecosystem and the pyramid of energy.	5+5
2	***Attempt any two of the following*** a) Explain the major causes and consequences of Deforestation. b) Explain why water is considered a precious natural resource? Describe any two major water conflicts. c) Explain photovoltaic solar cells and comment on their advantages & disadvantages. d) Write a note on wind energy and wind turbines. Explain the advantages and disadvantages of wind energy.	5+5
3	***Attempt any two of the following*** a) What is acid rain? What are the causes and effects of acid rain? b) What is photochemical smog? (Support your answer with theory, reactions and effects of photochemical smog). c) Explain biochemical oxygen demand (BOD) and chemical oxygen demand (COD) in water. What are the differences between them? d) What does thermal pollution mean? Explain the sources, effects, and control measures for thermal pollution.	5+5
4	***Multiple-choice questions*** **(The students need to write only one Right Choice/Correct answer in booklet)** 1) The Chipko movement was launched for.... a) Stopping deforestation of hill forests in U.P. b) Land protection in U.P. (Garhwal region) c) Wildlife Management in U.P. d) Saving Narmada river and its tributaries 2) Eutrophication results from.... a) Phosphate pollution of rivers and lakes b) Unchecked growth of aquatic weeds in water bodies c) Sewage and agricultural run-off provide plant nutrients in water d) All of the above 3) What is bad ozone? a) Ozone in the stratosphere b) A natural protective layer in the atmosphere c) A pollutant found in the troposphere d) A gas that absorbs harmful UV radiation	1*10

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	4) Which of these is an example of ex-situ conservation? a) National park b) Wildlife sanctuary c) Biosphere reserves d) Seed bank 5) What is the best definition of biodiversity? a) The variety of ecosystems in a region b) The number of species in a specific area c) The diversity of life forms on Earth, including species, genes, & ecosystems d) The genetic variation within a species 6) The primary goal of watershed development is… a) To create artificial lakes and dams b) To promote water conservation and management c) To increase the water supply and ensure it for urban areas d) To increase water supply for agriculture e) All of the above 7) Sustainable development aimed for… a) Meets the present's needs without compromising the ability of future generations to meet their own needs b) Balance between economic and social needs and the preservation of natural resources and ecosystems c) Reducing pollution, combating climate change, protecting biodiversity, and ensuring the wise use of natural resources for the long term d) All of the above 8) Which of the following statements is true regarding dissolved oxygen and water quality? a) High BOD and high DO indicate a healthy water body. b) Low DO and high BOD are indicators of poor water quality. c) DO and BOD are not good indicators of water quality. d) High DO is always a sign of pollution. 9) World Environment Day is celebrated every year on…. a) January 12 b) June 5 c) July 18 d) December 2 10) What is an ecosystem? a) A community of organisms interacting with the abiotic environment b) Only living organisms in a particular area c) A part of the biosphere with only plants d) Only non-living components of the environment	
5	a) Explain the objectives and importance of Air (Prevention and Control of Pollution) Act. What can an individual do to support the Act? b) Explain the principles of sustainable development. What are the major challenges involved in sustainable development?	5+5
6	*Attempt any two of the following* a) Write a note on: Greenhouse effect. b) What is the meaning of solid waste and solid waste management? Explain the effects of poor solid waste management and the different methods involved in solid waste disposal. c) Write a note on the use and exploitation of mineral resources. Also, explain the environmental issues associated with the extraction and use of mineral resources.	5+5

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